THE DARK SKY

Claude Game Build Specification

Roblox implementation brief • Story • Systems • Technical requirements • Milestones

A shared world. A shared responsibility.

Use this document as the build contract for Claude. Inspect before editing, build incrementally, and prove each feature works before moving on.

1. Purpose and instructions

This is the implementation brief for Claude to build The Dark Sky in Roblox Studio. Treat it as the source of truth unless the project owner approves a change. Start by inspecting the existing project and scripts. Do not overwrite working code or claim a feature works unless it has been implemented and tested.

- Work in small, testable increments; prioritize a playable vertical slice over a large unfinished world.
- Separate implemented features, proposed features, placeholders, and known limitations.
- Before broad refactors, explain the plan and files affected.
- After each milestone, report changed files, setup steps, tests, observed results, and remaining issues.
- If Claude cannot access Roblox Studio or the project files, state that limitation and provide exact import/setup/test instructions.

This brief synthesizes the supplied 43-page Dark Sky Roblox Experience Design Spec with the current project decisions: the public title is The Dark Sky, and the named protagonists are Aadi, Aditya, and Adrika.

2. Game summary and design pillars

Genre: spooky-fun cooperative environmental adventure with exploration, bicycle traversal, lighting puzzles, constellation encounters, and critical-thinking choices. Platform: Roblox. Intended audience: approximately grades 7–8, with a reading level accessible to younger players. Full intended experience: roughly 75–100 minutes; build and test it in milestones.

Core message: Responsible lighting, not darkness at any cost. Preserve appropriate lighting for safe paths, doors, stairs, crossings, emergency use, and legitimate security. Improve where light goes, how bright it is, its color, and when it operates.

Taglines: “A shared world. A shared responsibility.” “The night was never empty. We simply stopped seeing it.” Core rule: “Don't eliminate light. Make every light count.”

1. Learn by doing: observe a problem, gather evidence, make a change, compare results.
2. Balanced solutions: never reward switching off every light regardless of safety or use.
3. Visible consequences: fixes affect the Darkness Meter, sky, ambient sound, wildlife cues, and realm restoration.
4. Critical thinking: the friendly Moon sometimes gives misleading advice; players must verify claims.
5. Cooperation must work with chat disabled and in solo play.
6. Prioritize mobile accessibility, short dialogue, reduced motion, visual audio cues, and performance tiers.

3. Story, cast, and twist

Premise

In Maple Hollow, a fictional cozy suburban neighborhood, streets, courts, parking areas, and buildings glow at night while the sky is nearly empty. Three friends discover a Rift at a covered bridge leading into the Glarelands, a distorted mirror-world where wasted light has collected and corrupted the constellations. They ride bicycles, investigate with a Lux Lens, and restore the night by correcting lighting problems.

Protagonists

- Aadi — Explorer: finds clues and routes, notices details, and asks the questions that begin investigations.
- Aditya — Scientist: uses the Lux Lens, compares readings, tests hypotheses, and explains cause and effect.
- Adrika — Strategist: coordinates the team and weighs solutions against safety, environmental effects, and practical needs.

These are the named protagonists. The detailed reference also proposes Navigator, Tinkerer, and Naturalist gameplay roles. Map those roles to the named characters only with owner approval; do not assume unique abilities are already implemented.

The Moon twist

The Moon initially appears cheerful, funny, and helpful. Her motive is loneliness and jealousy: as human light drowned out the stars, she became the dominant bright object in the sky and began feeding on wasted glare. She subtly steers the friends toward incomplete or misleading conclusions. In the finale, she reveals herself as the true boss. The epilogue must clarify that natural moonlight and the real Moon are not the problem; the corrupted Moon spirit is.

- Foreshadow with a brief red glint in the Moon's eye and a slightly detuned chime.
- After each realm, the Moon grows subtly brighter even as the environment improves.
- Restored bosses deliver cryptic lines hinting someone else caused the corruption.
- The Skull Crusher, the penultimate boss, hints that the Moon created or manipulated the corruption.

Acts

1. Act I — The Missing Stars: meet the friends and Moon, learn bicycle controls, use the Lux Lens, audit three Home Street fixtures, and discover the Rift.
2. Act II — The Corrupted Constellations: explore five realms, solve lighting puzzles, calm bosses, collect evidence, and evaluate Moon tips.
3. Act III — The Night Returns: face the Skull Crusher as a five-principle recap, discover the Moon's deception, complete the Moon finale, watch the stars return, and finish the Night Sky Pledge.

4. World and realm map

Use one main Roblox place for Maple Hollow, the Glarelands, and Starlight Hub unless performance testing proves a separate finale place is necessary. Landmarks should be recognizable in both worlds; a fix in the Glarelands changes its matching place in Maple Hollow. Use fictional names for all published locations and do not import or trace Google Earth/reference imagery into the shipped game.

Maple Hollow location	Realm / boss	Learning focus
Home streets and cul-de-sacs	Ursa's Den — Ursa Major, Sleepless Bear	Light trespass into homes; sleep and rest
Main loop road and streetlights	Leo's Glare Gate — Blinded Lion	Shielding, aiming, and glare
Clubhouse, courts, pool floodlights	Orion's Spire — Orion, the Hunter	Skyglow/uplighting and timing after events
Creek trail and meadow edge	Scorpius Meadow — Poisoned Scorpion	Warm/amber light and habitat
Pond, fountain, dock	Cetus Pond — Cetus, the Stalker	Targeted waterside lighting and wildlife
School/church/library parking	Overglow Core — Skull Crusher	Unnecessary lighting and waste
Covered bridge	Rift / Eclipse Throne transition	Move between worlds
Barn-style clubhouse	Starlight Hub	Safe home base, telescope, pledge wall

The short concept originally had four constellation challenges; the supplied detailed design spec expands this to five by adding Ursa Major. Use the five-realm version here unless the owner explicitly changes scope.

5. Five principles of responsible lighting

Principle	In-game verb	Mechanic
Useful	QUESTION	Decide whether light is needed; protect safety-critical lighting.
Targeted	SHIELD / AIM	Shield fixtures and direct beams toward the intended area.
Low level	DIM	Use the lowest brightness that still provides suitable visibility.
Controlled	TIME	Use appropriate timers or motion controls; light only when needed.
Warm-colored	WARM	Prefer warm light (3000K or lower); consider amber near sensitive habitats.

Every puzzle, boss, fact card, and pledge item maps to one or more principles. Safety-critical lights at paths, doors, stairs, and crossings may be improved but cannot be switched off by puzzles. The intended solution is effective, responsible lighting—not total darkness.

6. Core loop and gameplay systems

Core loop

1. Explore the zone and find a fixture, clue, or environmental symptom.
2. Investigate with the Lux Lens and collect evidence/readings.
3. In co-op, compare reasoning using preset phrases or a team vote.
4. Use a fixture menu to QUESTION, SHIELD/AIM, DIM, TIME, or WARM.
5. Server validates the action and updates fixture and zone state.
6. Show before/after results; update stars, sky, wildlife cues, and audio.
7. Calm the boss, restore the constellation, award a Star Journal entry, then receive a Moon tip.

Darkness Meter

Maintain one authoritative 0–100 Darkness score per zone; higher means a healthier night environment. Define thresholds in config. Lighting, sky effects, audio, and wildlife cues derive from this single score. A fix should produce coordinated changes, not disconnected effects. Use thresholds such as 0/25/50/75/100 for gradual blending. Meters must use numbers/shapes as well as color.

Lux Lens and fixtures

- Lux Lens presents understandable approximations of brightness, glare, color temperature, light spill, and sky quality. If values are simulated, label them game estimates—not real measurements.
- Fixtures are tagged models with ProximityPrompt and large mobile-friendly targets.
- Interaction opens a radial or accessible list menu for the five verbs.
- Server validates player distance, action, allowed range, cooldown, puzzle state, and safety-critical restrictions.
- Fixture data includes shielded state, aim angle, brightness, color temperature, and schedule.
- Provide reset/undo for testing and explain invalid or unsafe actions.

Bicycles

Bicycles are the main traversal mechanic. Use constraint-based physics with responsive controls and server-side speed/position sanity checks. Each bike has a shielded, warm, downward headlamp. Restored dark zones may charge a Night Vision boost. Automatically dismount near puzzles.

- Mobile: auto-pedal, thumbstick steering, on-screen Brake / Boost / Bell.
- PC: WASD steering, Shift boost, Space brake; support gamepad.
- Bike travel must not block access to core puzzles; provide a recovery path if stuck.

Boss encounters

Bosses are original, large constellation creatures made from star points, Beams, particles, and a rigged body mesh. Use readable telegraphs and puzzle windows. Player penalty is temporary Glare (screen washout/slowdown), never gore or a death screen. Recover in safe shadow or via teammate high-five. Failed phases restart quickly at a checkpoint.

7. Boss requirements

Boss	Theme	Required mechanic / lesson
Leo, the Blinded Lion	Glare from unshielded fixtures	Shield and aim lights; create useful illumination and safe shadow lanes.
Orion, the Hunter	Skyglow, uplighting, and lights left on after activity	Aim uplights down; schedule or dim suitable lights after events.
Ursa Major, the Sleepless Bear	Light trespass into homes and disrupted rest	Reduce spill; dim appropriately; use timers where suitable.
Scorpius, the Poisoned Scorpion	Blue-rich light and ecosystem effects	Use warm/amber lighting and protect the meadow.
Cetus, the Stalker	Pond/fountain/dock lighting	Shield, redirect, or schedule appropriate lights; restore wildlife cues.
The Skull Crusher	Waste and over-lighting	Penultimate recap of all five lighting principles.
The Moon	Misinformation from a friendly source	Five-phase finale; use evidence to counter misleading tips.

The five realm bosses precede the Skull Crusher and Moon finale. The final boss must be completable even if the player earns zero Doubt Tokens.

8. “Doubt it!” critical-thinking mechanic

The Moon delivers 10 configurable tips: six misleading and four true in the baseline design. Players choose “Sounds right” or “Doubt it!” Correctly challenging a misleading tip earns a Doubt Token and Star Journal entry with evidence. Every misleading tip must be disproven by evidence available before the reveal.

- Each tip config includes text, isMisleading, evidenceKey, trigger, and follow-up dialogue.
- Tokens shorten or weaken Moon phases but are not required to win.
- No free-text response; choices use preset UI and work with chat disabled.
- Dialogue bubbles are at most 90 characters and roughly grade-6 reading level.
- Example misleading tip: “Just switch EVERY light off. Darkness is always best!” Evidence: safety lighting matters; use appropriate light, not no light.
- Fact cards and claims must be reviewed by the project owner before release.

9. UI, dialogue, journal, and pledge

Comic bubbles

- Four styles: speech, shout/taunt, thought, narration/system.
- Use BillboardGui for world dialogue and ScreenGui for narration/cinematic text.

- Support typewriter reveal, tap/click to complete, auto-advance, and queue up to three bubbles.
- Pause dialogue during boss attack telegraphs; offer reduced-motion/text presentation options.
- All player-facing text must come from configuration/localization, not hard-coded script strings.

Star Journal

Each realm entry records the problem, evidence, change, before/after outcome, and one reviewed real-world fact with plain-text source attribution.

Night Sky Pledge

At the end, players select at least one safe action for each category: Home, School, Community. Generate an in-game Star Card. Examples include asking an adult about timers/shielding, an adult-supervised school lighting audit, or reporting glare to the appropriate authority with an adult.

Never tell players to climb, unscrew, or rewire fixtures. Paths, stairs, doorways, and crossings must remain appropriately lit.

10. Technical architecture

Use Luau with --!strict and a service/controller pattern. Server owns progress, puzzle validation, boss state, party state, and persistence; clients handle input, presentation, local effects, and responsive bike feel.

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ReplicatedStorage/\n Config/ (ExperienceConfig, DialogueConfig, MoonTipsConfig, FactCardsConfig, ChecklistConfig, BossConfig,
FixtureConfig, Tuning, BubbleTheme, AudioConfig, FeatureFlags)\n Shared/ (Types, Signals, Util, Constants,
Net)\nServerScriptService/Services/ (GameState, Quest, BossAI, FixturePuzzle, Dialogue, Party, Data, Analytics,
Safety)\nStarterPlayer/StarterPlayerScripts/Controllers/ (Input, BubbleRenderer, HUD, Bike, LightingDirector, SkyFX, AudioDirector,
QualityScaler)\nWorkspace/ (tagged Zones and Fixture models)\ntests/ docs/ assets/ localization/
```

- Validate all RemoteEvents/Functions on the server: types, range, distance, cooldown, and state transitions.
- Clients never write progress directly. Use CollectionService tags for zones and fixtures.
- If persistence is implemented, use safe session-aware data handling for chapter flags, Doubt Tokens, Star Journal, and pledge choices.
- Store no names, emails, school names, precise locations, or unnecessary personal data.
- One main place and StreamingEnabled are the default; separate finale place is an explicit tested decision.

Configuration-first content

A non-programmer must be able to change title, dialogue, Moon tips, fact cards, pledge items, bubble theme, audio IDs, difficulty, and feature flags without editing gameplay scripts. Provide safe defaults, a validator, readable warnings, and an Admin Configuration Guide. Launch title is “The Dark Sky.”

11. Performance, accessibility, and safety

Support current iOS/Android phones, Windows laptops with integrated graphics, and gamepad. Targets: smooth 30+ FPS on supported mobile devices during fights and 60 FPS at default quality on suitable Windows laptops. Measure and report actual results; do not claim untested performance.

- Quality tiers Low/Mid/High control shadow lights, particles, post-processing, streaming radius, and audio simulation.
- Use LOD-friendly props and a practical light budget (design target around 12 shadow-casting lights in view at High).
- Dialogue is never voice-only; important sounds also have visual cues.
- Provide text-size, high-contrast, reduced-motion, and Story/Standard difficulty settings.
- No more than three flashes per second; no full-screen red flashes. Replace camera rolls/shakes with fades in reduced-motion mode.
- Meters must use shape/number as well as color.
- Core play works with chat disabled; use preset phrases/emotes; no free-text input in baseline.
- No ads, loot boxes, or paid game passes in baseline.
- Use original or properly licensed assets; disclose AI-generated assets and maintain a manifest.
- Use fictional place/institution names in the published game.

12. Visual and sound direction

Maple Hollow is a cozy suburban night gradually becoming strange. The Glarelands mirrors the same geography in cold blue-grey and near-black, cut by corrupted orange/red glows, fog, spores, distorted architecture, and original constellation creatures. Do not copy any recognizable franchise, character, music, logo, title, or design.

State	Audio	Visual
Polluted Maple Hollow	Electrical hum, distant traffic, buzzing fixtures	Warm skyglow, harsh glare, few stars
Restored Maple Hollow	Crickets, frogs, wind, owl/bat cues, warm piano motif	Clearer deep-blue sky, stars and wildlife return

State	Audio	Visual
Rift	Reversed bell, timber creaks, low swell	Covered-bridge world transition
Glarelands	Low drone, electrical crackle, original synth	Fog, spores, red storm glow, corrupted fixtures
Boss/restoration	Readable telegraphs, original boss motif, constellation lock chord	Clear attack lanes; vines/fog clear; stars return

Avoid disorienting camera effects. Effects must remain readable on mobile and have reduced-motion alternatives.

13. Analytics and learning

Use anonymous, aggregated analytics only: chapter funnel, realm completion, before/after results, Moon-tip accuracy, pledge selections, performance, and crashes. Do not record free-text chat or personal details.

Suggested 90-day targets from the reference are: at least 45% of starters reach the Pledge; a 25-point improvement on the three-question learning pulse quiz; and at least 70% correct calls in the final act. These are proposed goals, not proven results. Report actual sample size and outcomes.

14. Milestones

Milestone 0 — Audit

- Inspect existing Studio project, scripts, assets, and repository.
- Create an inventory marked working / partial / missing / placeholder / blocked.
- Confirm toolchain and missing assets. Do not start by generating the whole world.

Milestone 1 — Playable vertical slice (top priority)

1. Create a small Maple Hollow test area and one over-bright or poorly aimed fixture.
2. Implement one working fixture interaction and server validation.
3. Show Lux Lens evidence and a before/after panel.
4. Connect a zone Darkness Meter to visible sky/lighting feedback.
5. Add a simple Leo tutorial encounter/state machine and one Moon tip.
6. Test the complete flow in solo and two-player Studio sessions.

Milestone 2 — Core systems

- Configuration-driven dialogue and fixture data.
- Save/checkpoint/reset/rejoin behavior if persistence is enabled.
- Bike traversal and mobile/PC controls.
- Consistent zone score, sky/audio feedback, quality scaling.
- Preset co-op choices and Doubt It mechanic.

Milestone 3 — Full content

- Five realms and bosses: Leo, Orion, Ursa Major, Scorpius, Cetus.
- Skull Crusher recap and Moon reveal/five-phase finale.
- Starlight Hub, Star Journal, Night Sky Pledge.
- Localization-ready English and Spanish content.
- Original art/audio or approved licensed assets.

Milestone 4 — Release candidate

- Solo, two-player, and four-player playthroughs with chat disabled.
- Test mobile, Windows integrated graphics, and a discrete-GPU reference device.
- Test accessibility, saving/rejoin, server validation, and reset paths.
- Fix blocking/critical issues; document known issues and evidence.
- Publish only after owner approval of content, privacy, assets, and title.

Every milestone must deliver a playable build or clearly explain why it is not yet playable. The first priority is a reliable observe → change → compare → restore loop.

15. Definition of done

- A new player can complete the Leo vertical slice without developer help.
- Fixture changes are server-validated, replicated, and cannot switch off safety-critical lights.
- Darkness Meter, sky, audio, and wildlife cues respond consistently to zone score.
- Misleading Moon tips have evidence available before the reveal.
- Core progression works solo and with chat disabled.
- Mobile interactions use large targets and do not require keyboard input.
- Save/rejoin does not corrupt progression if persistence is implemented.
- Accessibility controls work.
- Config validator catches missing keys, overlong dialogue, invalid asset IDs, broken triggers, and missing translations.
- All assets are original/licensed and documented; no watermarks or third-party franchise IP.
- Test logs state platform, version/commit, pass/fail, known issues, and evidence.
- No blocking or critical defects remain at sign-off.

16. Repository and handover

Deliver a GitHub-ready source package and README that enables another person to clone, build, test, and publish the experience. Include project files, pinned tools/dependencies, config, localization, tests, scripts, asset manifest, and documentation.

- Use Rojo + Git if compatible; otherwise document the approved alternative.
- Use strict Luau, clear names, file headers, public-function comments, and commented config fields.
- Include formatter/linter/type-check/config-validation CI where feasible.
- Maintain assets/manifest.csv with source, author, license, and AI-generated status.
- Document architecture, data schema, RemoteEvent validation, analytics, runbook, admin config, and known issues.
- Never commit keys, passwords, webhook URLs, or secrets.
- Owner retains control of Roblox experience, source repository, and uploaded assets.

17. Rules for Claude's code changes

1. Summarize understanding and inspect existing files first.
2. List the next milestone and files to create/change before editing.
3. Implement only that milestone; avoid speculative refactors.
4. Provide complete files or patches and exact Roblox Studio/Rojo locations.
5. Give repeatable test steps and expected visible results.
6. State what is unimplemented, mocked, or untested.
7. Never claim a test was run unless it was actually run in the target environment.
8. Ask for Studio output, screenshots, or logs when needed to diagnose a failure.

18. First task for Claude

Begin with a project audit and a Leo vertical-slice plan. Do not attempt to build the entire 75–100 minute game in one pass.

1. Explain your understanding of The Dark Sky in your own words.
2. Inventory systems as existing / missing / uncertain based only on files actually inspected.
3. Propose a folder/module layout and identify required owner assets.
4. Define the first small implementation milestone and exact Studio test steps.
5. Then implement only the first milestone after confirming the project state.

Success condition for the first playable slice: a player inspects an over-bright or poorly aimed fixture, sees evidence, makes a safe adjustment, observes a meaningful before/after change, and restores Leo's realm. No safety-critical light may be switched off.